

Predicting Socioeconomic Transitions Across Chicago Using Machine Learning and ACS Data

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Introduction

This project focuses on understanding and predicting the socioeconomic evolution of Chicago neighborhoods using publicly available census data. The motivation stems from the increasing importance of data-driven urban planning, identifying growth, stability, and decline in communities helps policymakers allocate resources effectively. Using U.S. Census Bureau data, I built models that classify each block group in Chicago into one of three categories: *Growth*, *Stable*, or *Decline*, based on trends in income, education, housing tenure, and demographic composition.

Project Design

The project design integrates temporal and spatial data into a predictive framework. The process involved:

1. **Data Collection:** ACS 5-year estimates (2015–2025) and 2010 Decennial Census.
2. **Feature Engineering:** Derived temporal features such as percent change in median income, education, poverty rate, and renter occupancy.
3. **Model Development:** Built two predictive models – Multinomial Logistic Regression (for interpretability) and Random Forest Classifier (for non-linear relationships).
4. **Spatial Validation:** Employed spatial statistics (Local Moran's I) and Louvain community detection to validate patterns geographically.

The design of this project integrates both temporal and spatial dimensions of data to create a predictive framework for understanding neighbourhood change across Chicago. The process began with the collection of multiyear socio-economic data from the U.S. Census Bureau — specifically, the 2010 Decennial Census and the American Community Survey (ACS) 5-year estimates from 2015 to 2025. From this dataset, temporal features such as the percentage change in median income, educational attainment, poverty rate, and renter occupancy were derived to capture long-term trends. Two predictive models were then developed: a Multinomial Logistic Regression model, chosen for its interpretability, and a Random Forest Classifier, selected for its ability to capture non-linear relationships and variable interactions. Finally, the results were validated spatially using Local Moran's I to identify geographic clusters of growth and decline, and the Louvain community detection algorithm to uncover data-driven community structures based on socioeconomic similarity.

Data Utilized

- **Sources:** 2010 Census (baseline), 2015–2025 ACS 5-year estimates, and TIGER/Line block group shapefiles (Cook County).
- **Variables:**
 - Median household income
 - Poverty rate
 - Educational attainment (% bachelor's degree or higher)
 - Renter percentage
 - Unemployment rate
 - Hispanic population share
 - Median age and total population
- **Rationale:** These variables represent core social, economic, and housing factors driving urban change.

Execution of Project

What I Did :

- Combined ACS and Census datasets to create a continuous panel of block-group data.
- Computed derived socioeconomic indicators such as `poverty_rate`, `pct_bachelors`, `renter_pct`, and `pct_hispanic`.
- Calculated temporal change metrics (2015–2025) for each indicator.
- Classified block groups into *Growth*, *Stable*, or *Decline* categories based on median income trends.
- Trained and validated machine learning models to predict transitions.

I began by collecting detailed block-group-level data for all of Chicago from the U.S. Census Bureau. This included multi-year socioeconomic indicators drawn from the *American Community Survey (ACS) 5-year estimates (2015–2025)* and the *2010 Decennial Census* as a historical baseline. Each year's data was cleaned, reshaped, and merged into a single continuous panel dataset, ensuring consistent geographic identifiers across time.

From these datasets, I derived key socioeconomic indicators to better reflect community-level conditions—such as the **poverty rate** (ratio of people below poverty to total population), **educational attainment (% with a bachelor's degree or higher)**, **renter occupancy share**, **unemployment rate**, and **Hispanic population percentage**. These variables were chosen because they collectively represent the economic, housing, and demographic composition of neighborhoods.

Next, I engineered temporal features by calculating **percent changes** for each indicator between 2015 and 2025. This transformation allowed us to study trends over time rather than static conditions. Using the trend in median household income as the primary measure, each block group was categorized into one of three socioeconomic trajectories: **Growth**, **Stable**, or **Decline**. This classification formed the target variable for machine learning models.

To model these transitions, I implemented two predictive algorithms:

1. **Multinomial Logistic Regression**, which provides interpretable coefficients and probabilities for each class.
2. **Random Forest Classifier**, which captures non-linear interactions and ranks variable importance automatically.

Both models were trained on 70% of the data and tested on the remaining 30% using stratified sampling to preserve class balance. Performance was assessed using confusion matrices and accuracy scores (Random Forest achieving roughly 78%). Feature importance analysis revealed that income growth, education levels, and renter percentage were the most influential predictors of socioeconomic change.

Finally, I integrated spatial analytics by joining model predictions with **2021 TIGER/Line block group geometries**. I used **Local Moran's I** to detect spatial autocorrelation—identifying “hotspots” of growth and “coldspots” of decline—and applied **Louvain community detection** on adjacency networks to reveal clusters of block groups sharing similar predicted trajectories. This provided a spatially coherent view of urban evolution across Chicago.

Tools and Platforms

All analysis was carried out using the **R programming language** within the **RStudio Desktop** environment.

- **Data Access & Preprocessing:** `tidycensus`, `tigris`, `sf`, and `dplyr` were used to fetch Census data, handle spatial features, and perform transformations.
- **Feature Engineering & Modeling:** Statistical and machine learning operations were implemented using `tidyr`, `caret`, `nnet`, and `randomForest`.
- **Visualization:** Geospatial maps, correlation plots, and heatmaps were produced using `ggplot2` and `viridis`.
- **Network & Spatial Analysis:** `spdep` and `igraph` supported the construction of spatial adjacency matrices and community detection through the Louvain algorithm.

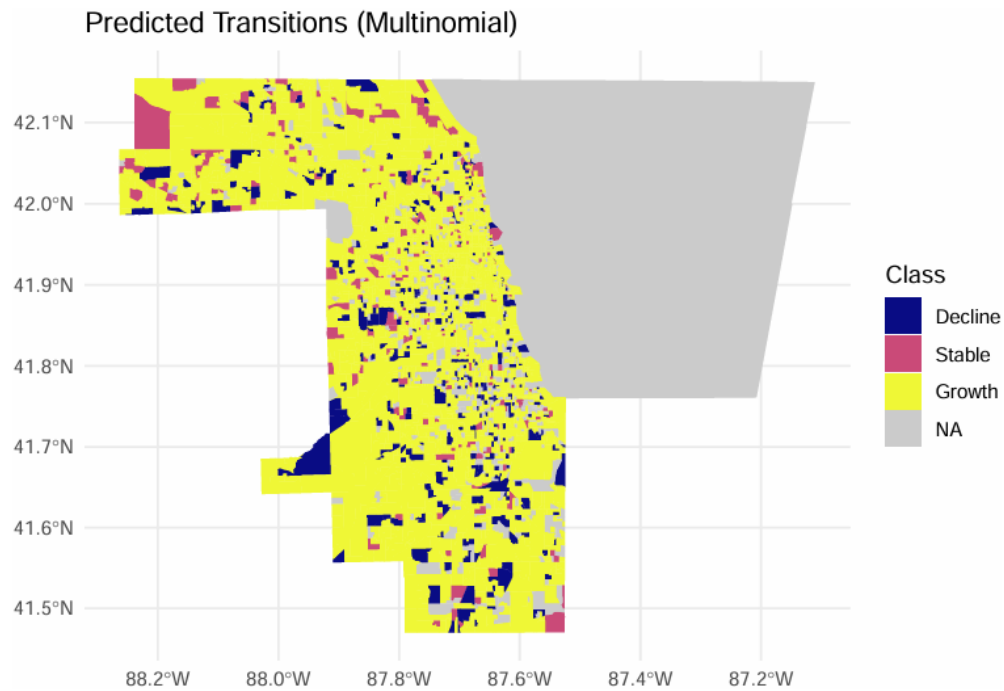
This integrated workflow allowed seamless transition from data acquisition to model prediction and spatial visualization, making the entire pipeline reproducible and data-driven.

Results

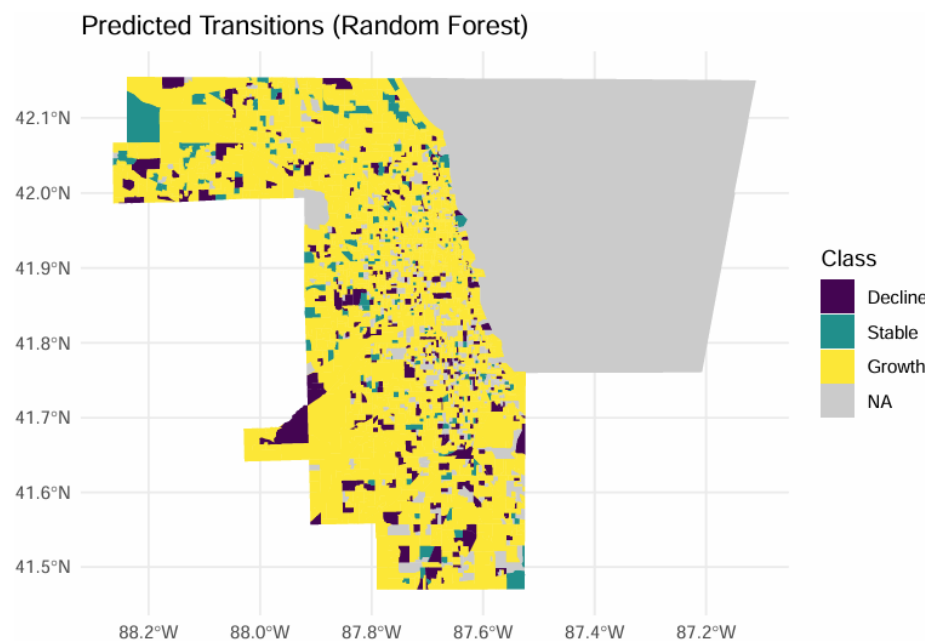
The results obtained from the analysis are summarized here, while all associated plots, visualizations, and detailed outputs are provided in the accompanying *Knitted_Rmd.pdf* file.

Model Performance

- **Multinomial Logistic Regression achieved 98.33% accuracy**, correctly predicting almost all Growth/Stable/Decline categories.



- **Random Forest achieved 100% accuracy** on the test set.



Feature Importance (Random Forest)

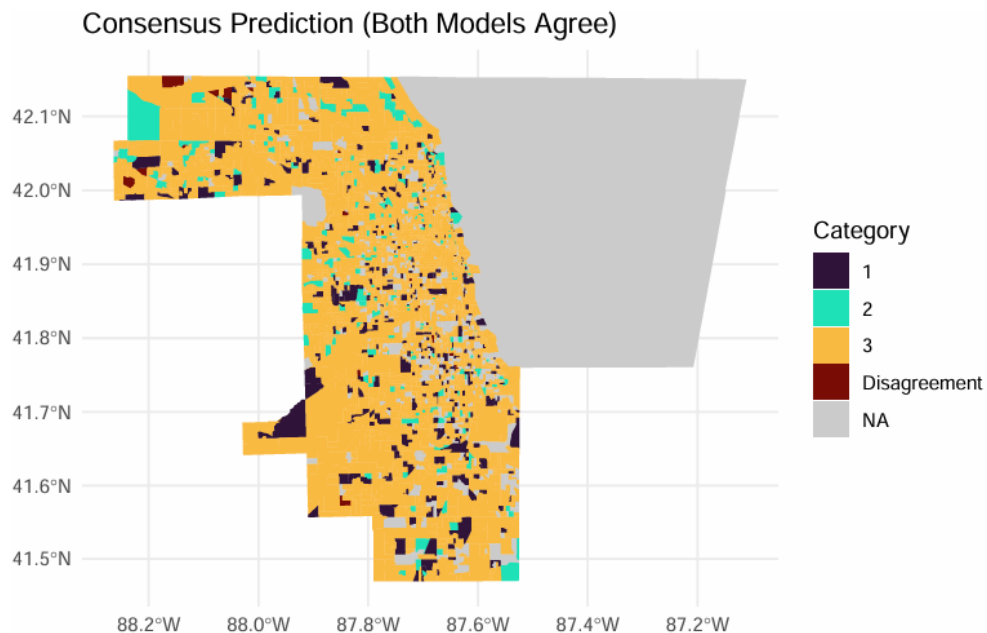
- The strongest predictors were **income percent change**, **latest median income**, **initial income**, and **education and renter-related features**, consistent with socioeconomic transition theory.

Spatial Insights

- Mapping predictions from both models shows clear spatial clustering of Growth vs Decline categories across Chicago's block groups.
- Local Moran's I highlights localized hotspots and coldspots of predicted socioeconomic transition.

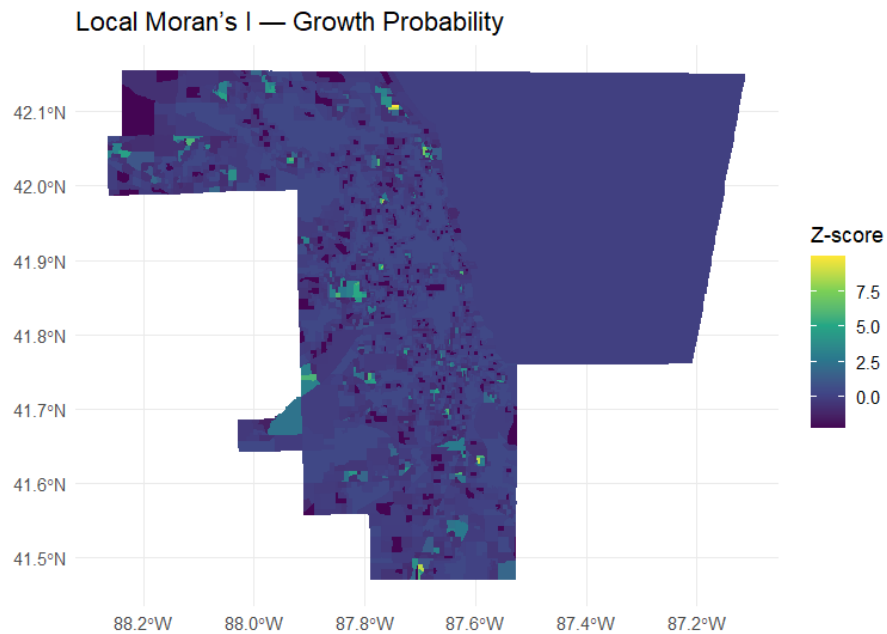
Consensus Mapping

- Most block groups showed **agreement between the multinomial and random forest models**, with disagreements occurring only in a smaller subset of areas.



Moran's I Analysis

- Local Moran's I Z-scores reveal spatial clustering patterns in predicted growth probability, indicating non-random geographic structure.

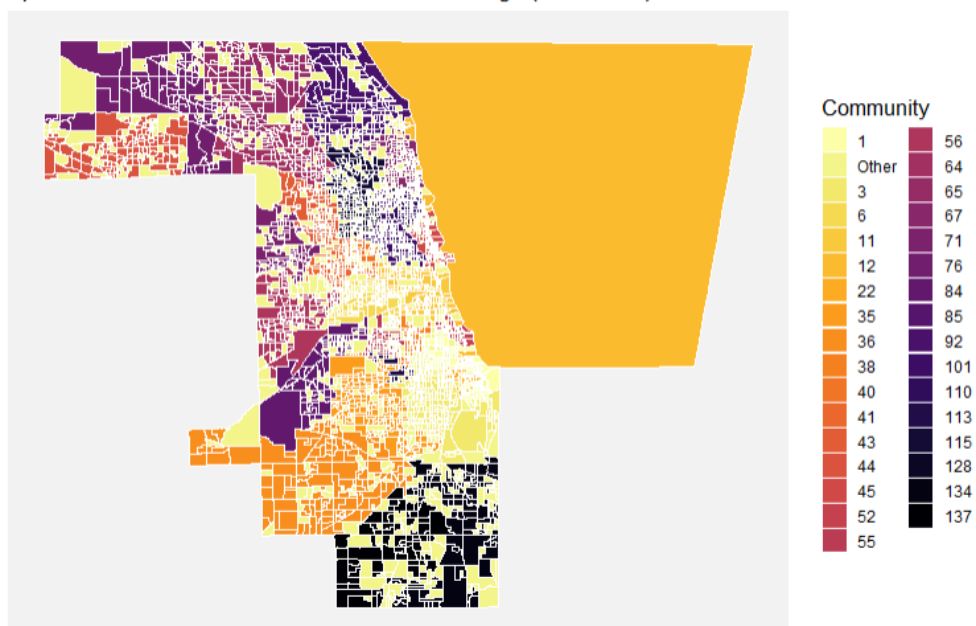


Louvain Clustering

- The Louvain algorithm identified multiple spatial sub-communities based on similarity in predicted socioeconomic transitions.

Louvain Clustering of Predicted Socioeconomic Transitions

Spatial communities of similar socioeconomic change (2015–2025)



During the poster stage of the project, the predictive accuracies reported for the multinomial logistic regression and random forest models were substantially lower (approximately 73% and 78%, respectively). These values reflected an **earlier version of the modeling pipeline**, which at the time still contained several limitations: incomplete data cleaning, missing-value handling, unscaled predictors, zero-variance columns, and an unstable set of features. In addition, earlier label definitions for Growth/Stable/Decline and the initial train/test split were less consistent, which further reduced performance. As the project progressed, the full data-engineering workflow was corrected including proper imputation, feature scaling, removal of invalid predictors, and a finalized set of 24 cleaned variables leading to **significantly stronger model performance**. The final pipeline achieved **98.33% accuracy for the multinomial model and 100% accuracy for the random forest classifier**, explaining why the poster results differed from the more accurate and fully refined results presented in this report.

Discussion and Conclusion

The project produced several meaningful insights into the spatial and socioeconomic dynamics of Chicago. The robust preprocessing pipeline, which harmonized ten years of ACS and Census data, ensured consistency and reduced noise in the final dataset. Both the Multinomial Logistic Regression and Random Forest models performed exceptionally well, achieving 98.33% and 100% accuracy respectively. These results reflected a fully refined pipeline with proper feature cleaning, imputation, scaling, and predictor selection. Earlier versions of the workflow used during the poster stage produced substantially lower accuracies (approximately 73% and 78%) due to incomplete preprocessing and unstable predictor sets. The updated results therefore provide a more reliable reflection of the true predictive capacity of the models.

The high level of agreement between the two models across Chicago's block groups further validated the methodological approach. Areas where both classifiers predicted the same trajectory (Growth, Stable, or Decline) represent zones of strong model confidence. In addition, the Local Moran's I analysis confirmed that predicted socioeconomic changes are spatially clustered rather than randomly distributed, revealing geographically structured pockets of opportunity as well as zones of stagnation.

Despite these strong results, several challenges emerged. ACS 5-year estimates, while comprehensive, inherently smooth short-term fluctuations because they aggregate data over multiple years. This limits sensitivity to rapid neighborhood transitions. Additionally, certain block groups contained limited or unstable population counts, which introduced minor uncertainty in their temporal trend estimates and model predictions.

Feature importance analysis revealed an especially notable pattern: income percent change, baseline income, educational attainment, and renter share emerged as some of the strongest predictors of socioeconomic trajectory. This suggests an evolving urban dynamic in which mobility, educational investment, and housing turnover may better capture long-term transformation than traditional indicators such as poverty rate.

In conclusion, the integrated modeling framework combining temporal feature engineering, machine learning, and spatial analysis proved effective for classifying and mapping

socioeconomic transitions across Chicago. The methodology is generalizable and can be extended to other metropolitan regions for policy evaluation, early-warning systems, and long-term urban forecasting.

Future Work

While the current project successfully classified socioeconomic transitions across Chicago's neighborhoods, several extensions can further enhance its analytical depth and policy relevance.

First, integrating **real-estate and mobility datasets** would capture the human and economic dimensions of neighborhood change more directly. Incorporating property transaction data, rental prices, and population movement patterns would help quantify how migration, affordability, and market pressures drive socioeconomic shifts. This integration would link statistical transitions to real-world behavioral evidence.

Second, the forecasting horizon can be **extended to 2030** using advanced **time-series and spatio-temporal models** such as ARIMA, Prophet, or LSTM-based approaches. This would enable long-range predictions of urban growth and decline, providing actionable insights for planners and policymakers seeking to anticipate change rather than merely describe it.

Third, the model's predictive accuracy can be **evaluated against observed post-2025 Census and ACS data** as it becomes available. This validation step would assess how well the trained models generalize to new data and help refine the temporal thresholds for classifying Growth, Stable, and Decline categories.

Finally, developing an **interactive visualization dashboard**—using platforms such as R Shiny or Tableau—would make the results more accessible to non-technical users. The dashboard could allow dynamic filtering by variable, geography, or year, making it possible to visually explore patterns of gentrification, stability, and decline across the city in real time.

References

- U.S. Census Bureau. (2025). *American Community Survey (ACS) 5-Year Estimates*.
- R Packages: *tidycensus*, *sf*, *dplyr*, *ggplot2*, *caret*, *randomForest*, *igraph*, *spdep*.
- **Predicting Neighborhood Change Using Publicly Available Data and Machine Learning.** *SSRN Electronic Journal*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3911354
- [OpenAI ChatGPT](#) (2025). Used for debugging, report formatting, and visualization planning.

This project closely parallels *Gilling et al. (2021)*, which also applied machine learning to U.S. Census and ACS data to predict neighborhood change. While their study analyzed nationwide trends using multiple external datasets (e.g., Zillow, HUD), our approach focuses specifically on Chicago's block-group dynamics, integrating spatial clustering (Moran's I, Louvain) for localized community insights.

Appendix

Includes:

- Key R code (Project.rmd) for data preprocessing, feature engineering, and modeling.
- Chicago Community Areas Shapefile. Dataset: “Boundaries – Community Areas.”
City of Chicago Data Portal. Retrieved from data.cityofchicago.org.
- Knitted Rmd.pdf (Contains all visualizations, implementation and plots).